

Last Time

Power

The Noncentrality Parameter

Determining Sample Size

Nonparametric Procedures

Bootstrapping

Wilcoxon Rank-Sum Test

Today

Nonparametric Procedures

- Bootstrapping

- Permutation Tests

- Wilcoxon Rank-Sum Test

- Wilcoxon's Matched-Pairs Signed-Ranks Test

Nonparametric Procedures

Sometimes none of the assumptions required for the t -test hold. Sometimes we don't know what the sampling distribution of a statistic is, or we don't want to estimate unknown parameters with sample statistics.

- ▶ Nonparametric or Distribution-free
 - ▶ Resampling
 - ▶ Rank tests (ranks)
 - ▶ Bootstrapping (raw scores)
 - ▶ Randomization tests
 - ▶ Permutation tests

The Population

- ▶ Bootstrapping for estimation: We don't know the population, but for these procedures we treat the data distribution as an estimate of the population distribution and construct sampling distributions by sampling from them.
- ▶ Permutation tests for p-values: These procedures *create* datasets where the null hypothesis or something equivalent to it is true by design.

With vs. Without Replacement

Bootstrapping: with replacement

Permutation: without replacement

Bootstrapping Example

Construct a 95% confidence interval for the *median* age of GSS respondents.

A large number of times:

- ▶ Sample N observations from a dataset with sample size N
- ▶ Compute the median for each new sample

Compute the 2.5% and 97.5% quantiles of the bootstrapped sample of medians.

Bootstrapping Example

```
> # Read in data
> data <- read.csv('GSS_1980.csv')
> age <- data$AGE
> N <- length(age) # sample size

> # Resample
> bignum <- 1000
> medians <- vector()
> for (i in 1:bignum)
>   medians[i] <- median(sample(age,N,replace=TRUE))

> # Compute the quantiles
> interval.95 <- quantile(medians,c(.025,.975))
```

```
2.5% 97.5%
 35     37
```

Two Independent Samples

Imagine each person's data in one of two boxes marked "Treatment" and "Control."
You know what the mean of the data in each box is.
Then oops! You drop the boxes and the data falls to the floor with no way of putting the observations back in the right box.

$$H_0 : F_X = F_Y$$

$$H_1 : F_X \neq F_Y$$

Two Independent Samples

Compute the mean difference of the original treatment and control groups. Let N_X and N_Y be the sample sizes for the treatment and control groups, respectively.

A large number of times:

- ▶ Select N_X observations from the scrambled data for a new treatment group, put the rest in the control.
- ▶ Compute the mean difference.

What is the percentile rank of the original mean difference with respect to the resampled distribution?

Two Independent Samples

```
> happy <- data$HAPPY < 3
> money <- data$REALRINC

> # Compute the mean difference
> obs.diff <- mean(money[happy]) - mean(money[!happy])

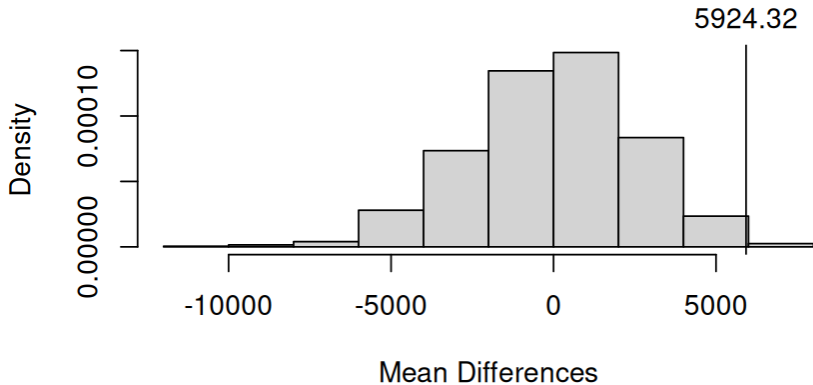
> bignum <- 1000
> N.happy <- sum(happy)
> N.total <- length(happy)

> # Resample mean differences
> diff <- vector()
> for (i in 1:bignum) {
>   choose <- sample(N.total, N.happy, replace=FALSE)
>   diff[i] <- mean(money[choose]) - mean(money[-choose])
> }

> # Percentile rank of obs.diff (upper-tailed test)
> cat("The percentile rank of ", round(obs.diff, 2), " is ", mean(diff < obs.diff),
>     ", which corresponds to a 'p-value' of ", mean(diff >= obs.diff))

The percentile rank of  5924.32  is  0.993 , which corresponds to a 'p-value' of  0.007.
```

Two Independent Samples



Dependent Samples

Permutation test: consider pair X_i, Y_i , e.g., pre- vs. post-treatment, where $X \sim F_X$ and $Y \sim F_Y$.

$$H_0 : F_X = F_Y$$

$$H_1 : F_X \neq F_Y$$

If H_0 is true, it shouldn't matter whether the X score or the Y score is assigned to the pre- or post-treatment condition (permute the observations).

Permutation Test

```
> data(anorexia)

> # Compute the difference scores
> obs.diff <- mean(anorexia$Postwt - anorexia$Prewt)
> N <- length(anorexia$Prewt)

> bignum <- 1000
> mean.diffs <- vector()
> for (i in 1:bignum) {
>   # Random vector of -1,+1
>   signs <- 2*rbinom(N,1,.5) - 1
>   mean.diffs[i] <-
>     mean(signs*(anorexia$Postwt - anorexia$Prewt))
> }

> # Percentile rank of obs.diff
> cat("The percentile rank of ",round(obs.diff,2)," is ",
>     mean(mean.diffs < obs.diff)," , which corresponds to a 'p-value' of ",
>     mean(mean.diffs >= obs.diff),".\n")

The percentile rank of 2.76 is 0.999 , which corresponds to a 'p-value' of 0.001 .
```

Permutation Test

Permutation test: consider pair X_i, Y_i , e.g., pre- vs. post-treatment, where $X \sim F_X$ and $Y \sim F_Y$.

$$H_0 : F_X = F_Y$$

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If H_0 is true, it shouldn't matter whether the X score or the Y score is assigned to the pre- or post-treatment condition (permute the observations).

More Explanation

- ▶ If there is no difference between the groups, then the differences we see pre- and post-treatment are just random variations. The direction of the differences (greater than or less than) are equally likely.

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- ▶ If the sign of the differences (positive or negative) is random (50/50), we can flip them back and forth without changing the characteristics of the distribution from which the observations were sampled.
- ▶ Reassigning the +s and -s to each pair a large number of times simulates random chance: pulling two measurements from a common distribution and calling one X and the other Y arbitrarily.
- ▶ This procedure *builds* a distribution of the differences that is consistent with the null distribution. The original observed difference can be compared to this distribution (p -value) to evaluate the null.

Step by Step

Patient	Pre-treatment	Post-treatment	Difference
1	80	85	5
2	90	88	-2
3	75	80	5
4	85	87	2
Mean	82.5	85	2.5

Step by Step

Patient	Pre-treatment	Post-treatment	Difference	Permute
1	80	85	5	-1
2	90	88	-2	1
3	75	80	5	1
4	85	87	2	-1
Mean	82.5	85	-1	

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Permutation Test

```
> data(anorexia)

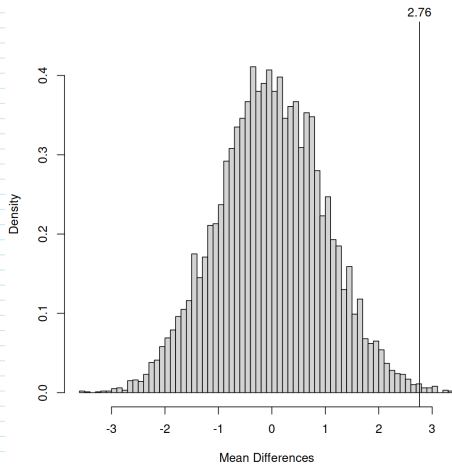
> # Compute the difference scores
> obs.diff <- mean(anorexia$Postwt - anorexia$Prewt)
> N <- length(anorexia$Prewt)

> bignum <- 1000
> mean.diffs <- vector()
> for (i in 1:bignum) {
>   # Random vector of -1,+1
>   signs <- 2*rbinom(N,1,.5) - 1
>   mean.diffs[i] <-
>     mean(signs*(anorexia$Postwt - anorexia$Prewt))
> }

> # Percentile rank of obs.diff
> cat("The percentile rank of ",round(obs.diff,2)," is ",
>     mean(mean.diffs < obs.diff)," , which corresponds to a 'p-value' of ",
>     mean(mean.diffs >= obs.diff),".\n")

The percentile rank of 2.76 is 0.999 , which corresponds to a 'p-value' of 0.001 .
```

Permutation Test



Distribution of $\bar{Y} - \bar{X}$ Under H_0

$$H_0 : F_X = F_Y$$

- ▶ Obviously $E(\bar{Y} - \bar{X}) = 0$ under H_0 .
- ▶ Under H_0 , all differences $Y - X$ are coming from a common distribution with mean $\mu = 0$ and variance estimated by s_D^2 .
- ▶ Estimate the (population) variance of $Y - X$ as $s_D^2 = \frac{\sum_{i=1}^N (D_i - \bar{D})^2}{N - 1}$

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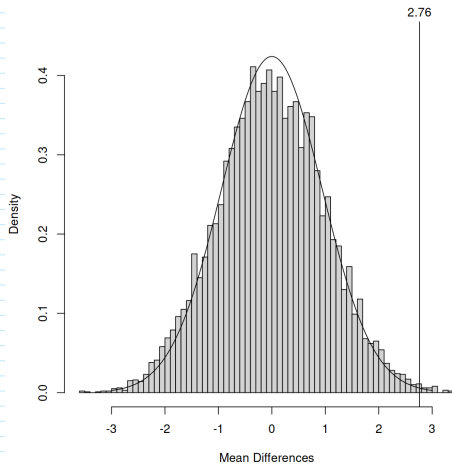
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- ▶ Estimate the (population) variance of $Y - X$ as $s_D^2 = \frac{\sum_{i=1}^N (D_i - \bar{D})^2}{N - 1}$
 - ▶ By the Central Limit Theorem, $\bar{Y} - \bar{X}$ comes from a normal distribution with mean $\mu = 0$.
 - ▶ The estimate of the variance of $\bar{Y} - \bar{X}$ is s_D^2/N .

$$N = 72$$

$$s_D^2 = 63.74$$

$$s_D^2 = 0.89$$

Permutation Test



Wilcoxon Rank Sum Test

$$H_0 : F_X(x) = F_Y(y)$$

$$H_1 : F_X(x) = F_Y(y - \theta)$$

- ▶ Consider $X = \{X_1, X_2, \dots, X_{N_X}\}$ and $Y = \{Y_1, Y_2, \dots, Y_{N_Y}\}$. If $\mu_X > \mu_Y$, if we rank-order all the scores in both groups from lowest to highest, the higher ranks should mostly be in group X and the lower ranks should be in group Y .
- ▶ If we then add those ranks, the sum of the ranks in group X should be higher than the sum of the ranks in group Y .
- ▶ If $F_X = F_Y$, then if we rank all the scores then the sum of the ranks for both groups should be about the same (modulo the sample size).
- ▶ Test statistic W_S is the sum of the ranks in the smaller group.

Example

$$X = \{70, 72, 47, 56, 61\}$$

$$Y = \{73, 90, 78, 83, 67, 65\}$$

$$\text{Ordered: } \{47, 56, 61, 65, 67, 70, 72, 73, 78, 83, 90\}$$

$$\text{Ranks: } \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$$

$$\text{Group: } \{1, 1, 1, 2, 2, 1, 1, 2, 2, 2, 2\}$$

$$W_S = 1 + 2 + 3 + 6 + 7 = 19$$

Example

$$X = \{70, 72, 47, 56, 61\}$$

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$$W_S = 1 + 2 + 3 + 6 + 7 = 19$$

- ▶ Smallest possible value of $W_S = 1 + 2 + 3 + 4 + 5 = \frac{N_X(N_X + 1)}{2} = 15$
- ▶ R transforms the Wilcoxon statistic W_S into the Mann-Whitney U .

$$U = W_S - \frac{N_X(N_X + 1)}{2} = 4$$

```
> 1 - pwilcox(4, 5, 6, lower.tail = FALSE)
```

```
[1] 0.02597403
```

GSS Survey

- ▶ Under the theory driving the (ersatz) birth order effect, only children (singletons) should feel more obligation toward their parents than children with siblings.
- ▶ Children who adhere to their parents' values may receive more support from their parents, resulting in greater opportunities, higher education, and ultimately higher-paying jobs.
- ▶ Hypothesis: Singletons who agree with the sentiment that younger people should be taught to respect the opinions of their elders may have higher salaries than those who disagree.

Mean Salaries:

Agree: \$26,141.33

Disagree: \$26,910.00

Median Salaries:

Agree: \$24,547.00

Disagree: \$15,294.50

Wilcoxon Rank Sum Test By Hand

```
> # Look at only children
> only <- data$SIBS==0

> # Subset the data
> groups <- data$YOUNGEN[only] # 1 is "agree"
> money <- data$REALRINC[only]
> N <- length(money)

> # Figure out which group has the smaller N
> smaller <- sum(groups==1) # Assume it's the agree group
> small.group <- 1
> # Check and swap to group 2 if it's smaller
> if (smaller > N/2) {
>   smaller <- N - smaller
>   small.group <- 2
> }

> # Order the ranks for all only children
> ranks <- rank(money,ties.method="average")

> # Compute the summed ranks and its p-value
> W.s <- sum(ranks[groups==small.group]) - smaller*(smaller+1)/2
> W.s

[1] 200

> 1 - pwilcox(W.s,smaller,N-smaller)

[1] 0.07936765
```

Using wilcox.test()

If sample sizes are large this can take a very long time or even crash.

```
> # Look at only children
> only <- data$SIBS==0

> # Subset the data
> groups <- data$YOUNGEN[only] # 1 is "agree"
> money <- data$REALRINC[only]

> wilcox.test(money~groups,alternative="greater",correct=FALSE,exact=FALSE)

Wilcoxon rank sum test

data: money by groups
W = 200, p-value = 0.07965
alternative hypothesis: true location shift is greater than 0
```


Using Resampling

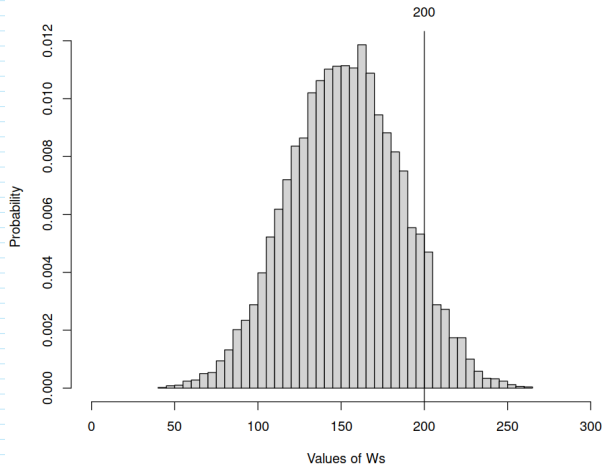
```
> bignum<- 10000
> W <- vector()
> # Select randomly from the integers 1:N
> for (i in 1:bignum) W[i] <- sum(sample(N,smaller,replace=FALSE))

> shift <- smaller*(smaller+1)/2 # Sum of 1:smaller
> lower <- 0
> upper <- sum((N - smaller + 1):N) - shift

> W <- W - shift
> hist(W,freq=FALSE,main="",breaks=50,
>       xlab="Values of Ws",ylab="Probability",xlim=c(lower,upper))
> abline(v=W.s)
> text(W.s,.0122,W.s,xpd=NA)
> mean(W>=W.s)

[1] 0.0882
```

Resampling Distribution of W_s



Revisit the First Example: Ties

$$X = \{70, 73, 47, 56, 61\}$$

$$Y = \{73, 90, 78, 83, 67, 61\}$$

$$\text{Ordered} \quad \{47, 56, 61, 61, 67, 70, 73, 73, 78, 83, 90\}$$

$$\text{Ranks} \quad \{1, 2, x, x, 5, 6, y, y, 9, 10, 11\}$$

$$\text{Group} \quad \{1, 1, 1, 2, 2, 1, 1, 2, 2, 2, 2\}$$

Revisit the First Example: Ties

X	$=$	$\{70, 73, 47, 56, 61\}$
Y	$=$	$\{73, 90, 78, 83, 67, 61\}$
Ordered		$\{47, 56, 61, 61, 67, 70, 73, 73, 78, 83, 90\}$
Ranks		$\{1, 2, x, x, 5, 6, y, y, 9, 10, 11\}$
Group		$\{1, 1, 1, 2, 2, 1, 1, 2, 2, 2, 2\}$

There are several alternatives to dealing with the tied ranks x and y .

- ▶ Replace x and y with the averages of the ranks (3.5 and 7.5; this changes the sampling distribution of W_S)
- ▶ Perform the test twice:
 - ▶ In the first test, assign the smaller of the ranks to Group 1 and the larger to Group 2. Compute the p -value.
 - ▶ In the second test, assign the larger of the ranks to Group 1 and the smaller to Group 2. Compute the p -value.
 - ▶ If the decision to reject or fail to reject H_0 changes, the data are inconclusive.

Wilcoxon Rank Sum vs. Mann-Whitney U Tests

- ▶ These are the same thing. The test is often called the Wilcoxon-Mann-Whitney test.
- ▶ The Mann-Whitney U statistic shifts the Wilcoxon statistic W_S down to 0.
- ▶ You can use the larger sample instead of the smaller, or just pick the value of U computed for both samples that is the smallest.

Matched Pairs

- ▶ While the rank sum test gives the nonparametric analog to the two independent samples t -test,
- ▶ the Wilcoxon signed-ranks test gives the nonparametric analog to the two dependent samples t -test.
- ▶ Rank the absolute differences for each group.

Suddath et al. 1990

Left hippocampal volume in identical twins:

Pair	Normal	Schizophrenic	Difference	Rank	Signed Rank
1	1.94	1.27	0.67	15	15
2	1.44	1.63	-0.18	9	-9
3	1.56	1.47	0.09	5	5
4	1.58	1.39	0.19	10	10
5	2.06	1.93	0.13	8	8
6	1.66	1.26	0.40	12	12
7	1.75	1.71	0.04	3	3
8	1.77	1.67	0.10	6	6
9	1.78	1.28	0.50	13	13
10	1.92	1.85	0.07	4	4
11	1.25	1.02	0.23	11	11
12	1.93	1.34	0.59	14	14
13	2.04	2.02	0.02	1	1
14	1.62	1.59	0.03	2	2
15	2.08	1.97	0.11	7	7

Suddath et al. 1990

Pair	Normal	Schizophrenic	Difference	Rank	Signed Rank
1	1.94	1.27	0.67	15	15
2	1.44	1.63	-0.18	9	-9
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$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
15	2.08	1.97	0.11	7	7

Let R_i be the signed rank for the i^{th} set of twins. Then the test statistic is:

$$T = \min \left\{ \left| \sum_{R_i > 0} R_i \right|, \left| \sum_{R_i < 0} R_i \right| \right\}$$

Suddath et al. 1990

Pair	Normal	Schizophrenic	Difference	Rank	Signed Rank
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Let R_i be the signed rank for the i^{th} set of twins. Then the test statistic is:

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- ▶ Let $T^+ = \sum_{R_i > 0} R_i = 15 + 5 + 10 + \dots = 111$.
- ▶ Let $T^- = \sum_{R_i < 0} R_i = -9$.
- ▶ Then $T = \min\{|111|, |-9|\} = 9$.

In R

```
> # Use the schizophrenia volume numbers
> twin.1 <- c(1.94,1.44,1.56,1.58,2.06,1.66,1.75,
>            1.77,1.78,1.92,1.25,1.93,2.04,1.62,
>            2.08)
> twin.s <- c(1.27,1.63,1.47,1.39,1.93,1.26,1.71,
>            1.67,1.28,1.85,1.02,1.34,2.02,1.59,
>            1.97)

> ##### By hand
> # Determine the sign of each difference
> signs <- sign(twin.1-twin.s)
> # and the ranks of the difference magnitudes
> ranks <- rank(abs(twin.1 - twin.s))

> # Compute the sums of the positive and negative ranks and determine the minimum
> T.pos <- sum(ranks[signs>0])
> T.neg <- sum(ranks[signs<0])
> T <- min(abs(c(T.pos,T.neg)))
> T

[1] 9

> # p-value (lower tail)
> psignrank(T,length(twin.1))

[1] 0.00100708
```

In R

```
> wilcox.test(x=twin.s,y=twin.1,alternative="less",  
>             paired=TRUE,exact=TRUE,correct=FALSE)
```

Wilcoxon signed rank exact test

data: twin.s and twin.1

$V = 9$, p-value = 0.001007

alternative hypothesis: true location shift is less than 0

Ties

- ▶ Define N = number of pairs (individuals).
- ▶ If $X_i - Y_i = 0$, eliminate the pair and reduce N by 1.
- ▶ If ranks are tied,
 - ▶ and both ranks have the same sign, break the tie however you want.
 - ▶ and both ranks have a different sign, assign tied ranks and proceed as usual.

Normal Approximation

- ▶ When N_1, N_2 get very large, the distribution of W_S is approximately normal. Ordering $N_1 + N_2$ observations becomes very difficult.
- ▶ Because

$$\begin{aligned} E(W_S) &= \mu_{W_S} = \frac{N_1(N_1 + N_2 + 1)}{2} \\ \text{Var}(W_S) &= \frac{2N_2\mu_{W_S}}{12} = \frac{N_1N_2(N_1 + N_2 + 1)}{12}. \end{aligned}$$

- ▶ So,

$$z_{W_S} = \frac{W_S - \mu_{W_S}}{\sigma_{W_S}} = \frac{W_S - \mu_{W_S}}{\sqrt{\frac{2N_2\mu_{W_S}}{12}}}.$$

R does a normal approximation unless you tell it not to.